FINAL-YEAR PROJECT

An Agentic Multi-LLM Answer-Validation System

Can a system of language models decide, on its own, whether an answer is trustworthy — with no human answer key to check against?

Generators LLaMA 3 · Mistral **Verifier** phi3 **Judge** qwen3.5:9b **Dataset** 5,000 questions

01 · THE PROBLEM

A model can't grade its own homework

1. Ask a language model a hard question and it answers fluently — **whether or not it's correct**, and it can't reliably tell you which.
2. Having a second model "check" the first fails quietly — checkers **share the same biases** and rubber-stamp plausible text.
3. In this domain there is **no ground-truth answer key** — so accuracy can't be measured directly. Every design choice must be justified another way.

02 · THE IDEA

Trust through independent corroboration

Instead of one model checking another, use an ensemble that cross-examines itself — and gate the output on how much that ensemble agrees.

> **Core principle:** two models generate, two more evaluate blindly, three external checks corroborate, and one confidence score decides whether to ship the answer or try again.

03 · ARCHITECTURE

Six stages, one confidence score

STAGE 1–2
Generate
LLaMA 3 + Mistral, independently

STAGE 3
Verify
phi3 scores both, blind

STAGE 5
Judge
qwen3.5:9b scores + picks winner

STAGE 4
Corroborate
Wikipedia · Wikidata · agreement

STAGE 6
Gate + Combine
score ≥ 0.72 or regenerate

The key control: a per-question coin flip randomizes answer order, and evaluators never see which model wrote which — so position and brand can't leak into a score.

04 · METHOD

Every choice earned its place

With no correctness labels, validity came from two families of label-free test — both of which can *fail*, which is what makes them real.

DISCRIMINATION

Can it tell bad from good?

A usable evaluator must score a deliberately broken answer far below a solid one. This test alone rejected two judge models.

CONSISTENCY & VALIDITY

Do the signals cohere?

Signals should predict each other, and a low confidence score should genuinely predict that a retry improves the answer.

05 · FINDING

Judge chosen on discrimination, not averages

JUDGE MODEL	MEAN / 80	FLAT-RATE	VERDICT
qwen2.5	71.9	—	Ceiling bias
CompassJudger-1	69.3	57.5%	No real analysis
qwen3.5:9b	**65.5**	**0.0%**	**Adopted**

CompassJudger-1 gave all eight rubric criteria the **identical score** on 57.5% of answers — a fluent number with no reasoning behind it. Similar averages hid a decisive difference visible only per-criterion.

06 · FINDING

Position bias was hiding as quality

Early on, one model was always shown first and prompts named the brands. Once evaluation was made blind and order-randomized, the apparent gap collapsed:

BY RAW POSITION (ARTEFACT)

35% · 65%

A vs B — inflated by the ordering

BY TRUE MODEL IDENTITY

45% · 55%

LLaMA 3 vs Mistral — the real, small gap

Separating the two proved most of the "gap" was an artefact — and quantified the modest part that was genuine.

07 · FINDING

The 0.72 threshold, proven three ways

METHOD	RESULT
Two-proportion z-test	p = 0.011 — the drop-off is real
ROC / Youden's J	optimal 0.70–0.74, AUC 0.63
2,000-sample bootstrap	**95% CI [0.69, 0.75]**

> **Solved by design:** a control group gave a random slice of already-passing answers a retry too — the only way to get unbiased data above the threshold.

08 · RIGOR

Three measurement bugs — caught, not shipped

1. A model's **self-reported total** didn't match its own scores → now always recomputed from the parts.
2. A "second-opinion" signal was **secretly derived from the judge**, inflating results → excluded from the search.
3. A threshold analysis **under-counted part of the data** after a sampling rate changed → re-weighted correctly.

Each was found by a result looking *too clean* and being checked — and corrected before it became a conclusion.

09 · HONEST LIMITS

What the evidence doesn't settle

The system prefers Mistral (55%). Is that real quality, or a judge quirk?

THE JUDGE SAYS

Mistral is more factual

Higher on correctness, faithfulness, hallucination-freedom.

THE VERIFIER SAYS

No difference at all

Same answers, zero gap on any criterion — and it's proven able to spot real gaps.

Reported as **open**, with the evidence for both sides — not forced to a tidy conclusion.

10 · DELIVERED

The project at a glance

5,000
questions, one consistent regime

8,432
attempts, every signal logged

6
pipeline stages, fully instrumented

35
dated research-log entries

A working pipeline, a bias-corrected dataset, a statistically calibrated confidence gate, and a transparent record of how every decision was justified.

IN ONE LINE

The contribution isn't a pipeline that runs — it's a method for trusting it where no ground truth exists.

Thank you Questions welcome